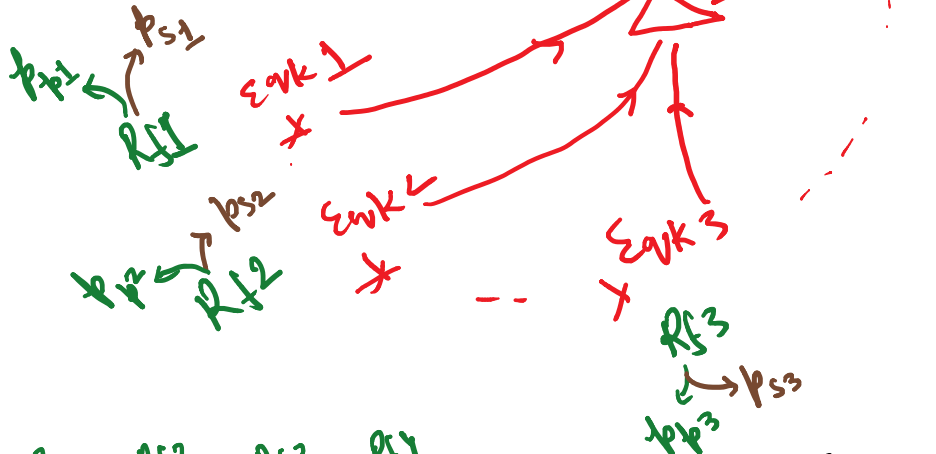


# Velocity Analysis of Rfs.



Time

Rf's

Gather

Rf1 Rf2 Rf3 Rf4

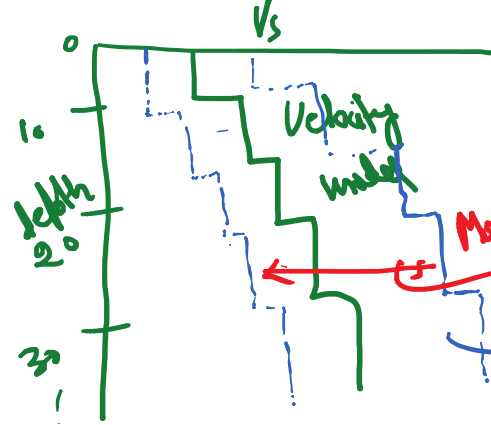
Master's Trace

This may be your reference earth model (aka 135° PREM, Litho, IASPEI, previous studies)

Velocity model

Time to depth domain

Optimization [SA, VFSA, NA, GA, PSO, GWO ---] (Iterative modelling) Global O.T.



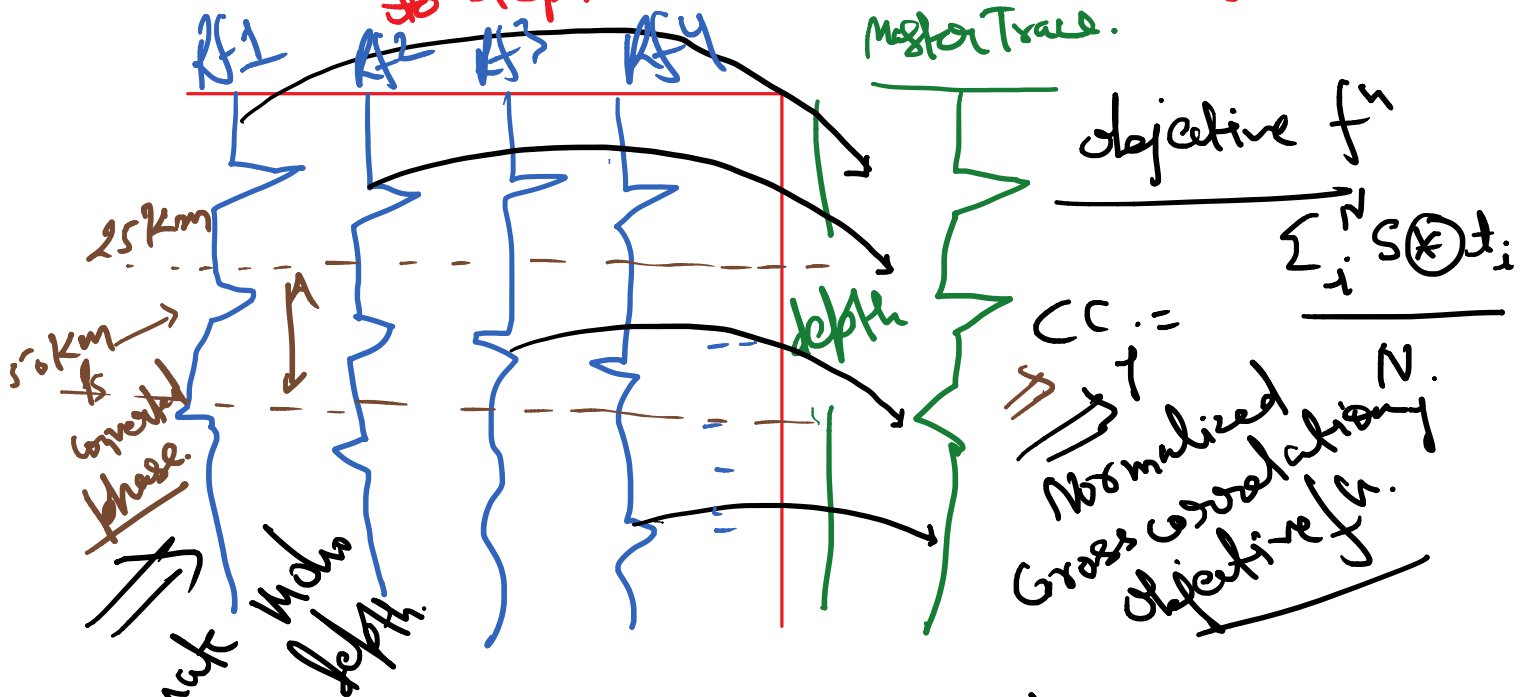
Model search space.

Model at each iteration will be searched to convert time domain receiver function to depth domain.

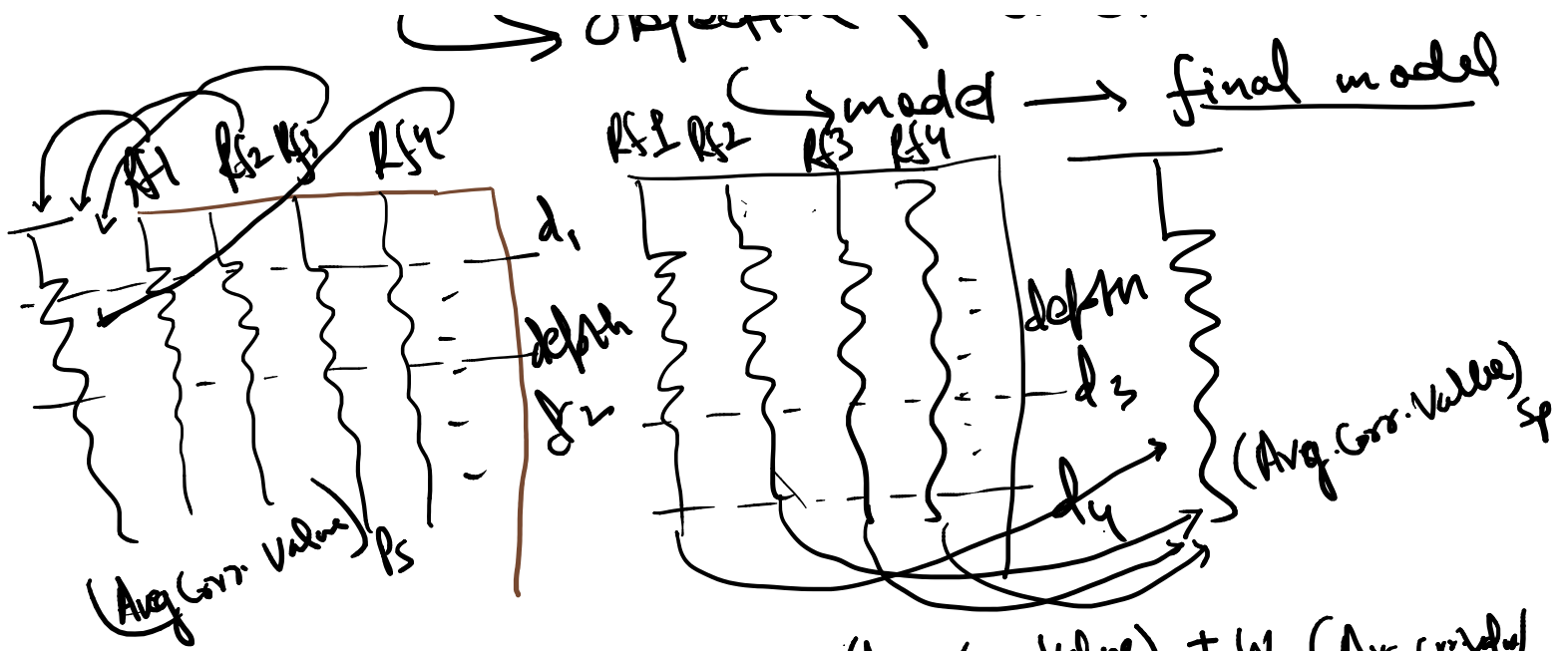
A hand-drawn blue curve, possibly representing a function or a path. A vertical blue line segment intersects the curve. A small red dot is located above the curve, to the right of the vertical line.

$$\Delta t = t_s - t_p = \sum_{i=1}^N z_i \left[ \sqrt{\frac{1}{\beta_i^2} - v_p^2} - \sqrt{\frac{1}{\alpha_i^2} - v_p^2} \right]$$

Major Trace.



(Aim)  
 Iter 1  $\rightarrow$   $CC_1$   $\xrightarrow{\text{Compared}}$   
 Iter 2  $\rightarrow$   $CC_2$   $\xrightarrow{CC_2 < CC_1}$  model accepted  
 Iter 3  $\rightarrow$   $CC_3$   $\xrightarrow{CC_3 < CC_2}$  model updated  
 \vdots  
 Until  
 \(\rightarrow\) objective fn will be converged.  
 \(\rightarrow\) model  $\rightarrow$  final model



$$\text{Joint Corr. Error} = W_{P_s} (\text{Avg. Corr. Value})_{P_s} + W_{S_p} (\text{Avg. Corr. Value})_{S_p}$$

$$W_{P_s} > W_{S_p}$$

$$W_{P_s} + W_{S_p} = 1$$

$W_{P_s}$  → weight assigned to  $P_s$  Rts  
 $W_{S_p}$  → weight assigned to  $S_p$  Rts

$$CC_j = W_{P_s} \left[ \frac{\sum_{i_p}^{N_p} S^P \otimes t_{i_p}^{i_p}}{N_K^P} \right] + W_{S_p} \left[ \frac{\sum_{i_s}^{N_s} S^S \otimes t_{i_s}^{i_s}}{N_K^S} \right]$$

Joint Correlation Error

Estimating the depth of  
 No. of LAB.